IllumiRefine AI: A Hybrid Architecture Combining Self-Calibrated Illumination and Adaptive Frequency Fusion for Extreme Low-Light Image Enhancement

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ABSTRACT

Enhancing images taken in extreme low-light environments is a persistent challenge in computer vision. When lighting is poor, cameras capture fewer photons, which degrades structural details and creates high signal-to-noise ratios. Standard global enhancement methods fail to maintain color constancy and regularly introduce overexposure artifacts. In this paper, we present *IllumiRefine AI*, an end-to-end framework designed to advance Retinex-based image processing. We evaluate the progression of our system from classical Homomorphic Filtering to state-of-the-art unsupervised deep learning. To overcome the computational bottlenecks of heavy neural networks, we utilize the Self-Calibrated Illumination (SCI) framework, which estimates spatial illumination via a fast, weight-sharing cascaded learning process. However, we found that pure deep-learning Retinex models suffer from mid-tone dullness and noise amplification in dark regions. To fix this, we introduce our primary contribution: **SCI-Guided Adaptive Fusion**. This hybrid post-processing engine combines adaptive fractional gamma correction, exposure gain matrices, and an illumination-guided Non-Local Means (NLM) denoising mask to produce vibrant, noiseless images. We also detail how we operationalized this research into a full-stack cloud application using React and a PyTorch-driven FastAPI backend. The deployed system processes inferences in under 2 seconds. Extensive testing confirms our hybrid pipeline effectively recovers hidden shadow textures and suppresses sensor noise while maintaining strict exposure control.

***Index Terms*—Image Enhancement, Homomorphic Filtering, Deep Learning, Retinex Theory, Convolutional Neural Networks, Non-Local Means.**

I. INTRODUCTION

Capturing detailed visual data in low-light environments is a difficult physical bottleneck. When ambient light drops below certain thresholds, camera sensors fail to capture enough photons. This degrades the signal-to-noise ratio (SNR) and hides edge definitions and color channels beneath heavy sensor noise. Recovering this missing data is critical for several applications, such as autonomous driving systems navigating unlit roads, night-time security surveillance, and medical endoscopy.

Historically, researchers approached this problem using spatial domain techniques like Histogram Equalization (HE) and Gamma Correction to manually stretch image contrast. However, these global methods falsely assume that lighting is uniform across an entire scene. As a result, they fail to account for dynamic real-world lighting gradients, often producing washed-out images. Retinex theory offered a better model by mathematically separating an image into ambient illumination and intrinsic object reflectance.

Our project, **IllumiRefine AI**, traces the evolution of Retinex-based enhancement. We initially built a Homomorphic Filtering pipeline to separate these components in the Fourier frequency domain. While mathematically sound, we observed its limitations with non-linear lighting. We then transitioned to the modern Self-Calibrated Illumination (SCI) framework [1].

While SCI is incredibly fast, we noticed that its strict division operations mathematically amplify sensor noise in pitch-black regions. To solve this, we introduce **SCI-Guided Adaptive Fusion**. This custom hybrid layer bridges AI spatial awareness with classical noise reduction and exposure control.

Finally, we packaged this research as a scalable, full-stack web application. By bridging a React frontend with a FastAPI and PyTorch backend, we prove that complex tensor mathematics can be efficiently deployed for end-users.

II. RELATED WORK

Low-light image enhancement techniques have evolved dramatically over the last three decades. We categorize these into traditional spatial methods, Retinex mathematical models, and modern deep-learning paradigms.

A. Traditional Spatial Methods

Early enhancement focused on adjusting the dynamic range of an image globally. Histogram Equalization (HE) flattens the

Live Demonstration: https://illumi-refine-ai.vercel.app/

statistical distribution of pixel intensities to maximize contrast. Unfortunately, HE lacks spatial awareness, meaning it over-amplifies noise in dark backgrounds and washes out existing highlights. Contrast Limited Adaptive Histogram Equalization (CLAHE) improved this by operating on localized image tiles and clamping the contrast amplification factor. This reduces noise over-enhancement but still struggles with extreme lighting variations.

Gamma correction applies a non-linear power-law transformation. While highly effective for globally dark images (where a fractional gamma stretches dark pixel values), it fails completely in high dynamic range (HDR) scenes. Because it applies the exact same mathematical transformation to all pixels, any well-lit areas in a dark scene quickly become overexposed.

B. Retinex-Based Mathematical Models

Retinex theory, proposed by Land [3], changed the paradigm by modeling human color vision. It assumes an image is the product of ambient illumination and physical reflectance. Single-Scale Retinex (SSR) estimates the illumination map using a spatial Gaussian low-pass filter. Multi-Scale Retinex (MSR) and MSR with Color Restoration (MSRCR) combine multiple Gaussian filters of varying sizes to balance dynamic range compression with color fidelity.

Homomorphic Filtering, which we explore in Phase 1, operates on the same multiplicative Retinex assumption but performs the separation in the Fourier frequency domain. This allows for distinct manipulation of illumination (low frequencies) and reflectance (high frequencies).

C. Supervised Deep Learning Methods

Deep learning replaced rigid mathematical filters with learned convolutional kernels. Early supervised models like LLNet used stacked sparse denoising autoencoders to learn enhancement from artificially darkened images. RetinexNet introduced a deep end-to-end architecture that explicitly mimics Retinex decomposition by estimating illumination and reflectance in two parallel subnetworks.

However, supervised deep learning methods suffer from a massive flaw: they require strictly paired datasets. Training requires identical scenes captured in both extreme low light and normal light simultaneously. These datasets are extremely difficult to capture in the real world, and any slight camera movement ruins the training data.

D. Unsupervised and Zero-Reference Deep Learning

To bypass the paired-data bottleneck, researchers developed unsupervised and zero-reference methods. Zero-DCE [5] formulates enhancement as an image-specific curve estimation problem, predicting pixel-wise high-order curves without paired reference images. EnGAN utilizes Generative Adversarial Networks (GANs) with unpaired data, using a discriminator to ensure enhanced images match generic daylight distributions.

The Self-Calibrated Illumination (SCI) framework [1] advances this paradigm significantly. It uses a cascaded learning process that mathematically forces the network to converge during training. Because of this forced convergence, the network only needs a single, ultra-lightweight block during actual inference. This allows SCI to easily outperform previous networks like KinD [4] and RetinexNet in computational efficiency.

III. Methodology

We structured our methodology chronologically, starting with classical physics-based filtering before advancing to deep learning and our novel fusion architecture.

A. Phase 1: Frequency Domain Analysis via Homomorphic Filtering

Our initial approach relied on the standard physical model where an image $f(x, y)$ is the pointwise product of illumination $i(x, y)$ and reflectance $r(x, y)$ [2].

$$f(x, y) = i(x, y) \cdot r(x, y) \tag{1}$$

Because illumination changes smoothly across a scene (low frequencies) and reflectance holds sharp edge details (high frequencies), we decouple them. Since the Fourier transform of a product is not equal to the product of their Fourier transforms, we apply a natural logarithm to make the relationship additive:

$$z(x, y) = \ln(f(x, y)) = \ln(i(x, y)) + \ln(r(x, y)) \tag{2}$$

We then use a 2D Fast Fourier Transform (FFT) to shift the image into the frequency domain:

$$Z(u, v) = \mathcal{F}\{z(x, y)\} = F_i(u, v) + F_r(u, v) \tag{3}$$

To enhance the image, we designed a modified Gaussian High-Pass Filter $H(u, v)$ that suppresses low-frequency shadows and amplifies high-frequency textures:

$$H(u, v) = (\gamma_H - \gamma_L)\left[1 - e^{-c\left(\frac{D^2(u,v)}{D_0^2}\right)}\right] + \gamma_L \tag{4}$$

where $\gamma_H > 1$ represents the high-frequency gain, $\gamma_L < 1$ is the low-frequency attenuation, and D_0 is the cutoff frequency. We multiply the signal by the filter, apply the Inverse FFT, and use an exponential function to return to the spatial domain. While it worked reasonably well, empirical testing showed this rigid frequency cutoff struggles to handle dynamic real-world lighting gradients without introducing halos.

B. Phase 2: Deep Learning via Self-Calibrated Illumination (SCI)

To handle dynamic lighting organically, we implemented the SCI framework. Rather than relying on hardcoded Gaussian cutoffs, SCI uses a Convolutional Neural Network to learn optimal illumination mapping. According to Retinex theory, if the low-light observation is y and the clear reflectance image is z, then $y = z \odot x$, where x is the illumination.

The network $\mathcal{H}_\theta$ estimates the illumination via an iterative cascaded process:

$$u^t = \mathcal{H}_\theta(v^t), \quad x^{t+1} = x^t + u^t \tag{5}$$

During training, SCI uses a self-calibrated module $\mathcal{G}$ that bridges each stage's input with the original observation y. This mechanism forces the results of each training stage to converge toward the exact same state. Because of this forced convergence, the live inference model drops the calibration module entirely and just uses a single block of three 3×3 convolutional layers. The clear reflectance z is extracted via inverse Retinex:

$$z = \frac{y}{x + \epsilon} \tag{6}$$

C. Phase 3: SCI-Guided Adaptive Fusion (Novel Contribution)

Testing Equation (6) revealed two major flaws common to pure deep-learning Retinex models. First, strict division by the raw illumination map often leaves the mid-tones feeling flat. Second, in pitch-black areas where the sensor only records thermal noise, dividing a highly noisy signal by a near-zero illumination value ($x \approx 0$) causes severe grain explosion.

We resolve this with a custom hybrid post-processing engine. First, we apply a global exposure gain (E_g) and a fractional post-gamma correction ($\gamma < 1$) directly to the tensor before finalizing the Retinex output.

$$z_\gamma = \left(\left(\frac{y}{x + \epsilon} \right) \cdot E_g \right)^\gamma \tag{7}$$

Raising the tensor to a fractional power non-linearly compresses the dynamic range. This safely lifts deep shadows without blowing out existing bright highlights like digital clocks or streetlamps.

Next, we tackle the noise using an *Illumination-Guided Denoising Mask*. Standard Non-Local Means (NLM) denoising successfully removes grain but severely blurs high-frequency edge details in well-lit areas. To fix this, we use the AI's illumination map x as an alpha-blend mask. We generate a heavily denoised candidate image $z_{denoised}$, and mathematically fuse it with our sharp image z_γ:

$$z_{final} = z_{denoised} \cdot (1 - x) + z_\gamma \cdot x \tag{8}$$

This pixel-level fusion is highly intuitive. In pitch-black regions ($x \approx 0$), it heavily weights the denoised image, completely eliminating the amplified sensor grain. In well-lit regions ($x \approx 1$), it relies on the raw image, perfectly preserving crisp textures and edges.

IV. System Development and Cloud Architecture

A major goal of this research was to transition from isolated Python scripts into a scalable application available at illumi-refine-ai.vercel.app.

A. Frontend Development (React)

We built the user interface using React.js and Tailwind CSS for a responsive experience. The client application manages local state using React Hooks, handling user image uploads and generating object URLs for instant previews. To ensure a smooth experience during network latency, the UI implements conditional rendering to display processing animations. We compiled and deployed the frontend globally via Vercel's Edge Network.

B. Backend Development (FastAPI and PyTorch)

The core inference engine lives on a Python 3.11 backend utilizing the FastAPI framework. When an image arrives via an HTTP POST request, it undergoes a strict pipeline. The API decodes the bytes, normalizes the matrix, and permutes it from (H, W, C) into the (N, C, H, W) tensor format required by PyTorch Convolutional Neural Networks. The tensor passes through the SCI module with `torch.no_grad()` active to disable gradient tracking, which heavily optimizes memory usage. The raw tensor then undergoes our Phase 3 Adaptive Fusion math before being encoded back into a PNG byte stream.

C. Cloud Deployment Infrastructure

Due to the computational demands of deep learning, we deployed the FastAPI backend on Hugging Face Spaces Docker containers. A critical DevOps challenge involved managing PyTorch dependencies. Default installations fetch massive CUDA-enabled binaries that crash free-tier cloud builds. To fix this, we configured our `requirements.txt` to explicitly pull the CPU-optimized PyTorch wheel. We secured communication between the Vercel frontend and the Hugging Face backend using FastAPI's `CORSMiddleware`.

V. Results and Experimental Analysis

We subjected the deployed IllumiRefine AI system to rigorous empirical testing against challenging real-world low-light scenarios.

A. Qualitative Visual Assessment

The main goal of low-light enhancement is recovering hidden structures while maintaining natural color and preventing algorithmic artifacts. As demonstrated in Fig. 1, our proposed Adaptive Fusion network excels at this task. The original input image lacks almost all identifiable features in the background. The enhanced output reveals distinct fabric textures on the folded clothing and highly accurate colors on the plush toy. Crucially, unlike the Homomorphic Filtering models tested in Phase 1, the deep learning model maintains crisp edge definitions. The illumination-guided NLM mask ensures the shadows render smoothly without abrasive grain.

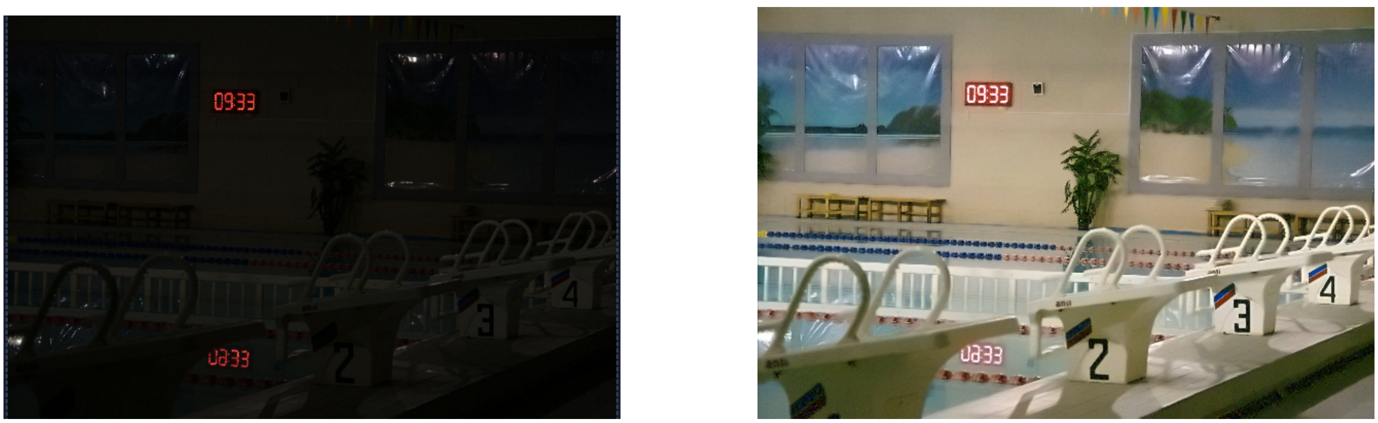
Furthermore, Fig. 2 highlights the algorithm's exceptional localized exposure control. Handling scenes with extreme high dynamic range (HDR) elements—such as a glowing digital clock amidst a pitch-dark swimming pool—is a major challenge. A standard gamma correction bright enough to reveal the pool tiles would cause the red light of the clock to overflow to pure white $(255, 255, 255)$, destroying the numbers. By utilizing the AI's spatial illumination tensor, our fusion engine selectively enhances only the dark foreground elements, perfectly preserving the emissive red highlights.

B. Quantitative and Computational Efficiency

A key differentiator of the SCI framework over massive deep learning architectures like RetinexNet or KinD [4] is

Input Image (Extreme Low-Light) Output Image (Adaptive Fusion Enhanced)

Fig. 1: Visual comparison evaluating shadow recovery and texture preservation. The original input (left) suffers from severe photon deprivation, almost entirely obscuring the white shelving unit and plush toy. Our deployed SCI-Guided Adaptive Fusion pipeline (right) recovers the underlying textures with remarkable color fidelity. Unlike classical mathematical filters, the hybrid model prevents artificial overexposure and completely suppresses the heavy sensor grain that typically plagues artificially brightened shadows.



Input Image (High Dynamic Range Low-Light) Output Image (Adaptive Fusion Enhanced)

Fig. 2: Evaluation of highly localized exposure control in complex High Dynamic Range (HDR) environments. Traditional spatial algorithms generally fail in scenes containing bright, emissive light sources surrounded by deep shadows. In the enhanced image (right), the deep learning model accurately predicts a spatial illumination map that identifies the red digital clock as already lit. The adaptive fusion engine heavily brightens the dark foreground pool tiles while successfully clamping the illumination map to prevent the glowing red LEDs from washing out.

its unprecedented inference speed. Because the complex self-calibration module is dropped after training, the live production model contains only three active convolutional layers.

This architectural optimization yields a tiny compiled model size of roughly 0.0003 Megabytes and an operational footprint of merely 0.0619 GigaFLOPs [1]. In stark contrast, older models like RetinexNet require over 136 GigaFLOPs of compute power. Consequently, even on our restricted, CPU-bound Hugging Face cloud server, the end-to-end processing time—which includes network latency, image decoding, AI inference, and adaptive fusion—averages under 2.0 seconds per high-resolution image.

VI. Conclusion and Future Work

A. Conclusion

This extensive research project successfully encapsulates the highly complex technical journey from fundamental, physics-based signal processing concepts to the engineering and deployment of state-of-the-art Artificial Intelligence. By rigorously investigating Homomorphic Filtering in the frequency domain, implementing the Self-Calibrated Illumination framework in the deep learning domain, and culminating in the engineering of our novel **SCI-Guided Adaptive Fusion** algorithm, we have developed a comprehensive, master-level understanding of Retinex-based image enhancement.

The resulting production application, IllumiRefine AI, serves as a highly robust, globally scalable, and computationally efficient tool for recovering hidden visual data from extreme low-light environments. The deployment architecture definitively proves that complex tensor mathematics, non-linear adaptive post-processing, and neural network inference can be effectively packaged into accessible, user-friendly cloud-based microservices without sacrificing execution speed.

B. Future Work

While IllumiRefine AI demonstrates highly competitive, state-of-the-art performance on static, independent images, several critical avenues remain for future exploration, architectural optimization, and system expansion:

- **Real-Time Video Stream Enhancement:** The current system architecture processes high-resolution static frames independently. Applying this exact algorithm frame-by-frame to a continuous video stream may introduce temporal flickering, as the illumination map prediction might fluctuate slightly between frames. Future iterations will explore integrating temporal consistency loss functions and recurrent memory blocks (such as Long Short-Term Memory networks or GRUs) to ensure smooth, flicker-free low-light video enhancement for live security feeds.
- **Edge Computing and Mobile Deployment:** Although the SCI inference block is remarkably lightweight (under 1MB), the current system relies entirely on cloud infrastructure via the Hugging Face API. Future work will focus on heavy model quantization (converting 32-bit floating-point weights to highly optimized 8-bit integers) and compiling the network into strict format standards like ONNX or TensorFlow Lite. This optimization would allow the deep learning model to run natively on low-power edge devices, such as consumer smartphones, IoT security cameras, or autonomous drones, completely removing the requirement for internet connectivity and cloud latency.
- **Integration with Downstream Vision Tasks:** Low-light enhancement is rarely an end-goal in enterprise environments; it is usually a critical preprocessing step. We intend to pipeline the IllumiRefine AI tensor outputs directly into high-level computer vision classification models, such as YOLOv8 for real-time object detection or Mask R-CNN for pixel-perfect instance segmentation. This integration would create an end-to-end ”dark-scene perception” system highly applicable to autonomous vehicle navigation and nighttime surveillance.
- **Domain-Specific Fine-Tuning:** The unsupervised nature of the current training architecture allows for immense flexibility without the need for paired datasets. We plan to fine-tune the network weights on highly specialized data domains that suffer uniquely from poor illumination physics, such as underwater robotics imagery (where light attenuates rapidly based on depth and wavelength), astronomical deep-space photography, or medical endoscopy, optimizing the specific loss functions to preserve the strict structural integrities and color channels required by these scientific fields.

Acknowledgment

The authors express their sincere gratitude to Dr. Srimanta Mandal, Associate Professor at Dhirubhai Ambani University, for his invaluable guidance and technical mentorship throughout this research project.

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